

Joint PDF factorization

From a generative model to local factors.

Point and extended objects.

Known and unknown object counts.

GrBP: Precluster, then associate whole groups.

S1

Point · known

S2

Point · unknown

S3

Extended · known

S4

Extended · unknown

Two modeling choices shape the graph.

Object model: one measurement or several per scan.

Object count: fixed, or represented by potential objects and existence variables.

	Known count	Unknown count
Point	<u>S1 · exclusivity</u>	<u>S2 · existence + new objects</u>
Extended	<u>S3 · Poisson counts</u>	<u>S4 · anchored new objects</u>

GrBP · Preclustering → group association

Fixed groups and an unknown target count

Prior $\times$ allocation $\times$ measurements

1. Read the counts from the association.
2. Insert the Poisson count laws.
3. Cancel allocation factorials.
4. Collect constants; marginalize auxiliaries.

$$f(S, A, Z) = f(S) p(A, m \mid S) f(Z \mid A, m, S)$$

S collects object states; A collects associations and any existence variables needed for the allocation.

What is observed, and what is inferred?

Observed: the ordered measurement vector Z and its count.

Inferred: states, associations, and existence when applicable.

Clutter parameters and measurement models are supplied.

point: $\mathbf{x}$ extended: $\mathbf{y} = (\mathbf{x}, \mathbf{e})$

uncertain existence: $r \in \{0, 1\}$

$$C(Z, m) = \frac{\mu_c^m e^{-\mu_c}}{m!} \prod_{j=1}^m f_c(\mathbf{z}_j)$$

For fixed data and clutter model, $C(Z, m)$ is constant in the inferred states and assignments.

Each target makes at most one claim.

n_t existing point targets; m measurements.

A nonzero claim must be unique.
Several targets may have the null claim.

$$a_i \in \{0, 1, \dots, m\}, \quad a_i = 0 : \text{missed detection}$$

$$\psi(\mathbf{a}) = \begin{cases} 1, & a_i \neq a_j \text{ whenever } i \neq j, a_i, a_j \neq 0, \\ 0, & \text{otherwise.} \end{cases}$$

$$\mathcal{D} = \{i : a_i \neq 0\}, \quad n_c = m - |\mathcal{D}|$$

Divide by the clutter explanation.

Claimed measurements use the target likelihood.

Unclaimed measurements use f_c .

Multiplying and dividing by clutter densities leaves a ratio for each detection.

$$\begin{aligned} f(Z \mid \mathbf{a}, \underline{\mathbf{X}}, m) &= \prod_{i \in \mathcal{D}} f(\mathbf{z}_{a_i} \mid \mathbf{x}_i) \prod_{j \in \mathcal{C}} f_c(\mathbf{z}_j) \\ &= \left[\prod_{i \in \mathcal{D}} \frac{f(\mathbf{z}_{a_i} \mid \mathbf{x}_i)}{f_c(\mathbf{z}_{a_i})} \right] \prod_{j=1}^m f_c(\mathbf{z}_j). \end{aligned}$$

A permutation removes the clutter factorial.

Given the detected targets, all consistent assignments to distinct measurement slots are uniform.

The clutter Poisson factorial cancels the allocation factorial.

$${}_m P_{m-n_c} = \frac{m!}{n_c!},$$

$$\frac{\mu_c^{n_c} e^{-\mu_c}}{n_c!} \frac{n_c!}{m!} = \frac{\mu_c^m e^{-\mu_c}}{m!} \prod_{i \in \mathcal{D}} \frac{1}{\mu_c}.$$

One factor $1/\mu_c$ remains for each detected target.

Detection ratios and missed detections

Each target contributes its prior and one local factor g .

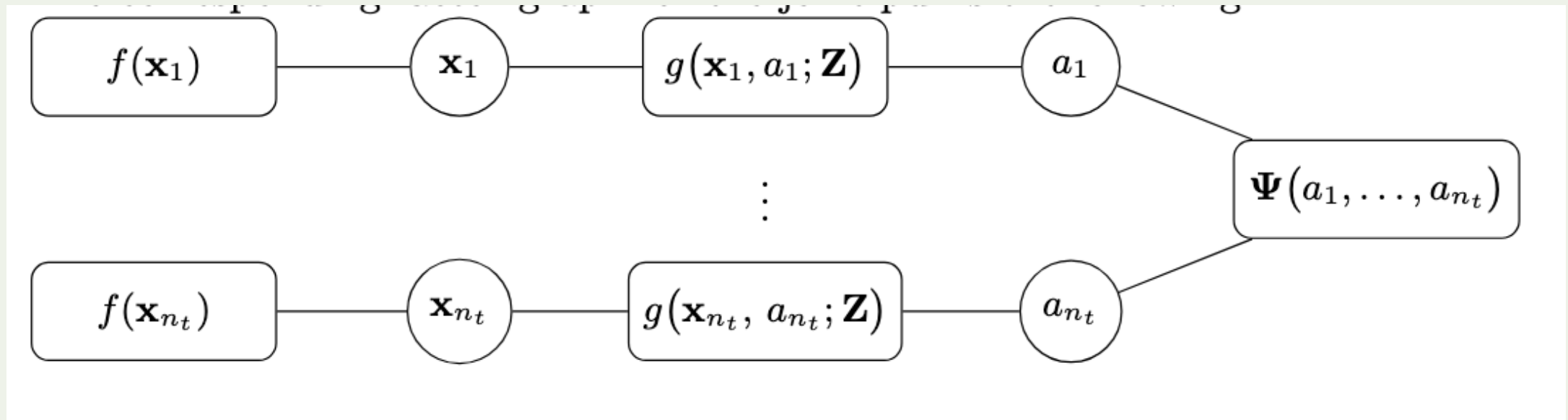
The consistency factor couples their association choices.

$$g(\mathbf{x}_i, a_i; Z) = \begin{cases} \frac{P_d(\mathbf{x}_i) f(\mathbf{z}_{a_i} | \mathbf{x}_i)}{\mu_c f_c(\mathbf{z}_{a_i})}, & a_i \neq 0, \\ 1 - P_d(\mathbf{x}_i), & a_i = 0. \end{cases}$$

$$f(\underline{\mathbf{X}}, \mathbf{a} | Z) \propto \psi(\mathbf{a}) \prod_{i=1}^{n_t} f(\mathbf{x}_i) g(\mathbf{x}_i, a_i; Z)$$

One chain per target; one shared constraint.

Prior $\rightarrow$ state $\rightarrow$ local likelihood $\rightarrow$ association. Ψ forbids duplicate nonzero claims.



Original Scenario 1 factor graph from the restored article.

Augment each state with existence.

Legacy potential objects carry

$$\underline{\mathbf{y}}_i = (\underline{\mathbf{x}}_i, \underline{\mathbf{r}}_i).$$

An existing object can be detected or missed. A nonexistent object cannot claim a measurement.

$$\underline{\mathbf{g}}(\underline{\mathbf{x}}_i, \underline{\mathbf{r}}_i, a_i; Z) = \begin{cases} \frac{P_d(\underline{\mathbf{x}}_i) f(\mathbf{z}_{a_i} | \underline{\mathbf{x}}_i)}{\mu_c f_c(\mathbf{z}_{a_i})}, & a_i \neq 0, \underline{\mathbf{r}}_i = 1, \\ 1 - P_d(\underline{\mathbf{x}}_i), & a_i = 0, \underline{\mathbf{r}}_i = 1, \\ 0, & a_i \neq 0, \underline{\mathbf{r}}_i = 0, \\ 1, & a_i = 0, \underline{\mathbf{r}}_i = 0. \end{cases}$$

Split unclaimed points into new objects and clutter.

Each measurement has a new potential object $\bar{\mathbf{y}}_j$.

On the valid support, new-object and clutter counts follow directly from existence and association.

$$\mathcal{N} = \{j : \bar{r}_j = 1\}, \quad n_n = |\mathcal{N}|, \quad n_c = m - |\mathcal{D}| - n_n$$

$$\binom{n_c + n_n}{n_n}^{-1} ({}_m P_{m-n_c-n_n})^{-1} = \frac{n_n! n_c!}{m!},$$

$$\mu_n^{n_n} \mu_c^{n_c} = \mu_c^m \left(\frac{\mu_n}{\mu_c} \right)^{n_n} \prod_{i \in \mathcal{D}} \frac{1}{\mu_c}.$$

A claimed point cannot also start a new object.

$\mathcal{T}$ contains points already claimed by legacy objects.

A valid new-object branch needs both the birth intensity μ_n and state density f_n .

$$\bar{g}(\bar{\mathbf{x}}_j, \bar{r}_j, \mathbf{a}; \mathbf{z}_j) = \begin{cases} \frac{\mu_n f_n(\bar{\mathbf{x}}_j) f(\mathbf{z}_j | \bar{\mathbf{x}}_j)}{\mu_c f_c(\mathbf{z}_j)}, & j \notin \mathcal{T}, \bar{r}_j = 1, \\ 0, & j \in \mathcal{T}, \bar{r}_j = 1, \\ f_d(\bar{\mathbf{x}}_j), & \bar{r}_j = 0. \end{cases}$$

Legacy chains plus new-object factors

The same $\psi(\mathbf{a})$ enforces unique legacy claims.

Each new-object factor depends on the entire legacy association vector.

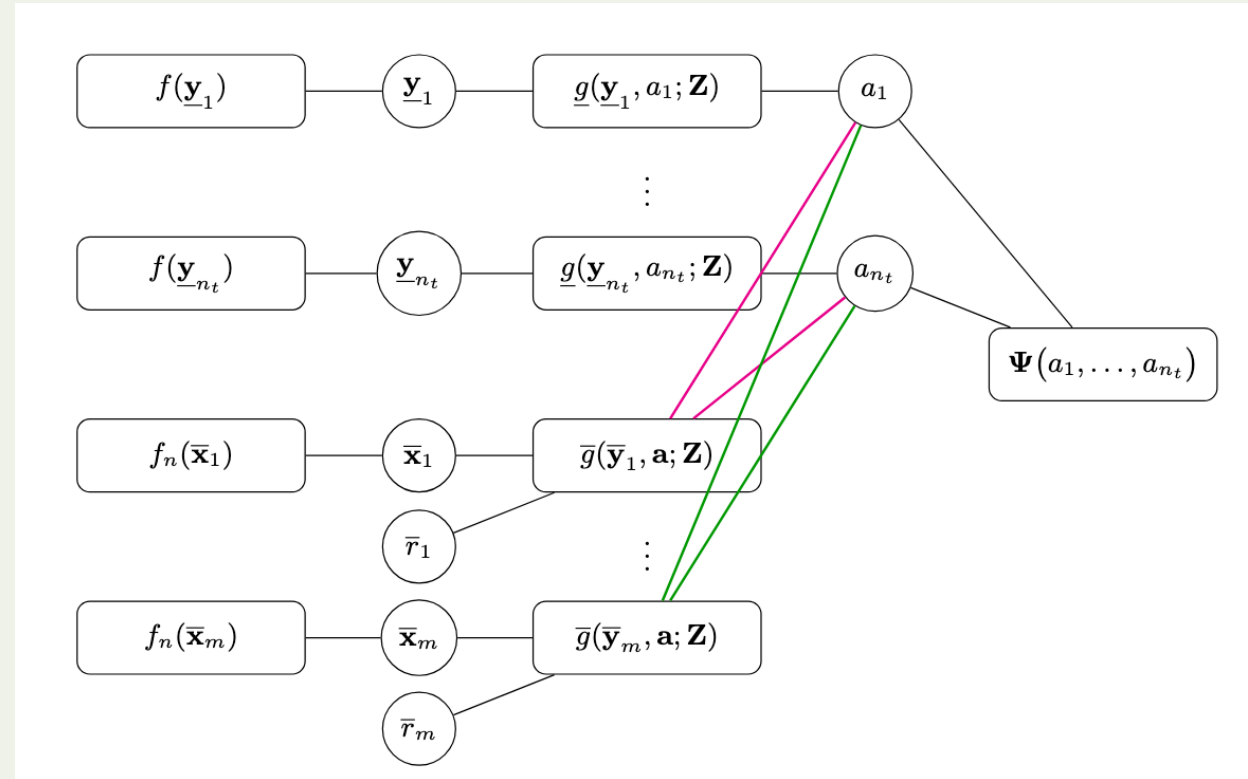
$$f(\underline{\mathbf{Y}}, \overline{\mathbf{Y}}, \mathbf{a} \mid Z) \propto \psi(\mathbf{a}) \prod_{i=1}^{n_t} f(\underline{\mathbf{y}}_i) \underline{\mathbf{g}}(\underline{\mathbf{y}}_i, a_i; Z) \\ \times \prod_{j=1}^m \bar{g}(\overline{\mathbf{y}}_j, \mathbf{a}; \mathbf{z}_j).$$

New candidates share the association decisions.

Legacy chains remain on the left.

New candidates add $\bar{g}$ factors.

The colored edges prevent a legacy claim and a new-object claim from explaining the same point.



Original Scenario 2 graph.

Let every measurement name its source.

The extended state is

$$\mathbf{y} = (\mathbf{x}, \mathbf{e}).$$

Several measurements can share one object. Each measurement has exactly one source; $b_i = 0$ means clutter.

$$b_i \in \{0, 1, \dots, K\}, \quad \mathcal{T}^k = \{i : b_i = k\}$$

$$p(|\mathcal{T}^k| \mid \mathbf{y}_k) = \frac{\mu_m(\mathbf{y}_k)^{|\mathcal{T}^k|}}{|\mathcal{T}^k|!} e^{-\mu_m(\mathbf{y}_k)}$$

The allocation binomials telescope.

Each object selects its cell from the remaining measurements.

The allocation contributes a factorial for each cell, cancelling the Poisson factorials.

$$\prod_{k=1}^K \binom{n_c + \sum_{k'=k}^K |\mathcal{T}^{k'}|}{|\mathcal{T}^k|}^{-1} = \frac{n_c! \prod_{k=1}^K |\mathcal{T}^k|!}{m!},$$

$$\mu_c^{n_c} \prod_k \mu_m(\mathbf{y}_k)^{|\mathcal{T}^k|} = \mu_c^m \prod_k \left(\frac{\mu_m(\mathbf{y}_k)}{\mu_c} \right)^{|\mathcal{T}^k|}.$$

One unary factor, then one edge per point

The zero-count exponential belongs to the object's unary factor.

An assigned point contributes a measurement-rate likelihood ratio. Every other edge is neutral.

$$g(\mathbf{y}_k) = f(\mathbf{y}_k) e^{-\mu_m(\mathbf{y}_k)}$$

$$g_{ki}(\mathbf{y}_k, b_i; \mathbf{z}_i) = \begin{cases} \frac{\mu_m(\mathbf{y}_k) f(\mathbf{z}_i | \mathbf{y}_k)}{\mu_c f_c(\mathbf{z}_i)}, & b_i = k, \\ 1, & b_i \neq k. \end{cases}$$

Put all edges inside the object product.

Each object contributes its unary and all m pairwise factors.

All objects meet at the measurement-oriented association variables.

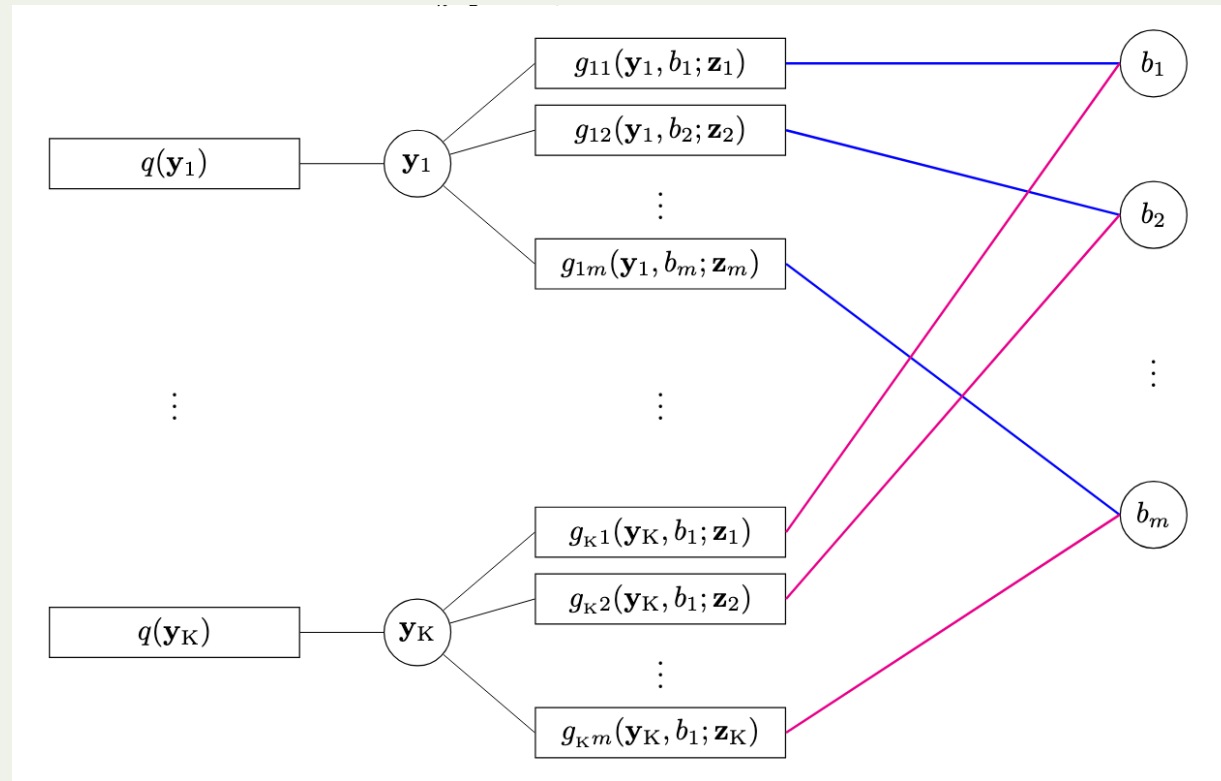
$$f(Y, \mathbf{b} \mid Z) \propto \prod_{k=1}^K \left[g(\mathbf{y}_k) \prod_{i=1}^m g_{ki}(\mathbf{y}_k, b_i; \mathbf{z}_i) \right]$$

Many measurements may connect to the same object.

Each object–measurement pair has a factor g_{ki} .

Variable b_i selects the source of measurement i .

The source choice itself gives each measurement one owner.



Original Scenario 3 graph.

Anchor each new object to its first point.

$\underline{K}$ legacy potential objects and
 M new potential objects.

New candidate k is anchored at
 measurement k and can explain
 later measurements.

$$b_i \in \{0, 1, \dots, \underline{K} + i\},$$

$$\bar{r}_k = 1 \iff b_k = \underline{K} + k,$$

$$\bar{l}_k = \sum_{i=1}^M \mathbf{1}\{b_i = \underline{K} + k\}.$$

A newly detected object has a nonzero count.

Newly detected objects use a zero-truncated Poisson count.

Existing legacy objects use an ordinary Poisson count;
nonexistent ones generate zero points.

$$p(l \mid \mathbf{x}, \mathbf{e}, l \geq 1) = \frac{\mu_m(\mathbf{x}, \mathbf{e})^l}{l!} \frac{e^{-\mu_m(\mathbf{x}, \mathbf{e})}}{1 - e^{-\mu_m(\mathbf{x}, \mathbf{e})}}$$

$$p(l \mid \mathbf{y}) = \begin{cases} \frac{\mu_m(\mathbf{x}, \mathbf{e})^l}{l!} e^{-\mu_m(\mathbf{x}, \mathbf{e})}, & r = 1, \\ \mathbf{1}\{l = 0\}, & r = 0. \end{cases}$$

Expand first, then remove the auxiliaries.

1. Introduce newly detected states and their associations.
2. Map them onto measurement-anchored candidates.
3. Integrate copied states; sum auxiliary labels and mappings.

$\tilde{\mathbf{y}}$: newly detected object states,

$\tilde{\mathbf{b}}$: their measurement assignments,

$\pi(k)$: new PO $k \mapsto$ newly detected object.

Relabeling cancels the new-object factorial.

Permuting the auxiliary labels of $\overline{K}$ newly detected objects gives the same canonical assignment. The sum contributes $\overline{K}!$.

$$\frac{\overline{K}! l_{\text{fa}}! \prod l_k!}{M!} \frac{\mu_n^{\overline{K}} e^{-\mu_n}}{\overline{K}!} \frac{\mu_{\text{fa}}^{l_{\text{fa}}} e^{-\mu_{\text{fa}}}}{l_{\text{fa}}!} \prod \frac{1}{l_k!} = \frac{\mu_n^{\overline{K}} \mu_{\text{fa}}^{l_{\text{fa}}} e^{-(\mu_n + \mu_{\text{fa}})}}{M!}$$

$$\overline{K} = |\mathcal{D}_b|, \quad l_{\text{fa}} = M - \sum_{k=1}^{\overline{K}} l_k - \sum_{k=1}^M \bar{l}_k$$

The count model is complete; values come next.

$\mathcal{D}_b$: existing new candidates. $\mathcal{D}_y$: existing legacy objects.

$$\begin{aligned}
 f(\bar{\mathbf{y}}, \mathbf{b}, M \mid \underline{\mathbf{y}}) &= \frac{\mu_n^{|\mathcal{D}_b|} \mu_{\text{fa}}^{l_{\text{fa}}} e^{-(\mu_n + \mu_{\text{fa}})}}{M!} \\
 &\times \prod_{k \in \mathcal{D}_b} \left[\bar{r}_k f_n(\bar{\mathbf{x}}_k, \bar{\mathbf{e}}_k) \mu_m(\bar{\mathbf{x}}_k, \bar{\mathbf{e}}_k)^{\bar{l}_k} \frac{e^{-\mu_m(\bar{\mathbf{x}}_k, \bar{\mathbf{e}}_k)}}{1 - e^{-\mu_m(\bar{\mathbf{x}}_k, \bar{\mathbf{e}}_k)}} \right] \\
 &\times \prod_{k \notin \mathcal{D}_b} \left[(1 - \bar{r}_k) f_d(\bar{\mathbf{x}}_k, \bar{\mathbf{e}}_k) \mathbf{1}\{\bar{l}_k = 0\} \right] \\
 &\times \prod_{k \in \mathcal{D}_y} \left[\mu_m(\underline{\mathbf{x}}_k, \underline{\mathbf{e}}_k)^{l_k} e^{-\mu_m(\underline{\mathbf{x}}_k, \underline{\mathbf{e}}_k)} \right] \prod_{k \notin \mathcal{D}_y} \mathbf{1}\{l_k = 0\}.
 \end{aligned}$$

Separate legacy prediction and new-object evidence.

$f^-(\underline{\mathbf{y}})$ includes the legacy existence mass and its conditional state density.

The new-object unary carries the detected-object intensity and the truncated-count normalization.

$$\underline{q}(\underline{\mathbf{y}}) = \begin{cases} f^-(\underline{\mathbf{y}})e^{-\mu_m(\underline{\mathbf{x}}, \underline{\mathbf{e}})}, & \underline{\mathbf{r}} = 1, \\ f^-(\underline{\mathbf{y}}), & \underline{\mathbf{r}} = 0. \end{cases}$$

$$\bar{q}(\bar{\mathbf{y}}) = \begin{cases} \mu_n f_n(\bar{\mathbf{x}}, \bar{\mathbf{e}}) \frac{e^{-\mu_m(\bar{\mathbf{x}}, \bar{\mathbf{e}})}}{1 - e^{-\mu_m(\bar{\mathbf{x}}, \bar{\mathbf{e}})}}, & \bar{\mathbf{r}} = 1, \\ f_d(\bar{\mathbf{x}}, \bar{\mathbf{e}}), & \bar{\mathbf{r}} = 0. \end{cases}$$

An absent object cannot own a point.

Write the rate-weighted likelihood ratio as ρ_{ki} .

Assignment to an existing object activates that ratio; an absent object forbids the assignment.

$$\rho_{ki} = \frac{\mu_m(\mathbf{x}_k, \mathbf{e}_k) f(\mathbf{z}_i \mid \mathbf{x}_k, \mathbf{e}_k)}{\mu_{\text{fa}} f_{\text{fa}}(\mathbf{z}_i)}$$

$$\underline{g}_{ki}(\underline{\mathbf{y}}_k, b_i; \mathbf{z}_i) = \begin{cases} \rho_{ki}, & b_i = k, \underline{\mathbf{r}}_k = 1, \\ 1, & b_i \neq k, \underline{\mathbf{r}}_k = 1, \\ \mathbf{1}\{b_i \neq k\}, & \underline{\mathbf{r}}_k = 0. \end{cases}$$

The anchor enforces existence in both directions.

If candidate k exists,
measurement k must belong to it.

If it does not exist, the anchor
cannot name it.

$$\bar{g}_k(\bar{\mathbf{y}}_k, b_k; \mathbf{z}_k) = \begin{cases} \rho_{kk}, & b_k = \underline{K} + k, \bar{r}_k = 1, \\ 0, & b_k \neq \underline{K} + k, \bar{r}_k = 1, \\ \mathbf{1}\{b_k \neq \underline{K} + k\}, & \bar{r}_k = 0. \end{cases}$$

An extended new object needs more than its anchor.

For every $i > k$, candidate k can explain measurement i through $\bar{h}_{ki}$.

Earlier measurements cannot refer to a later candidate.

$$\bar{h}_{ki}(\bar{\mathbf{y}}_k, b_i; \mathbf{z}_i) = \begin{cases} \rho_{ki}, & b_i = \underline{K} + k, \bar{r}_k = 1, \\ 1, & b_i \neq \underline{K} + k, \bar{r}_k = 1, \quad i > k \\ \mathbf{1}\{b_i \neq \underline{K} + k\}, & \bar{r}_k = 0. \end{cases}$$

Two families of local products

Legacy objects connect to every measurement. New candidates connect to their anchor and all later measurements.

$$f(\underline{\mathbf{y}}, \overline{\mathbf{y}}, \mathbf{b} \mid \mathbf{Z}) \propto \prod_{k=1}^K \left[\underline{q}(\underline{\mathbf{y}}_k) \prod_{i=1}^M \underline{g}_{ki}(\underline{\mathbf{y}}_k, b_i; \mathbf{z}_i) \right] \\ \times \prod_{k=1}^M \left[\bar{q}(\overline{\mathbf{y}}_k) \bar{g}_k(\overline{\mathbf{y}}_k, b_k; \mathbf{z}_k) \prod_{i=k+1}^M \bar{h}_{ki}(\overline{\mathbf{y}}_k, b_i; \mathbf{z}_i) \right].$$

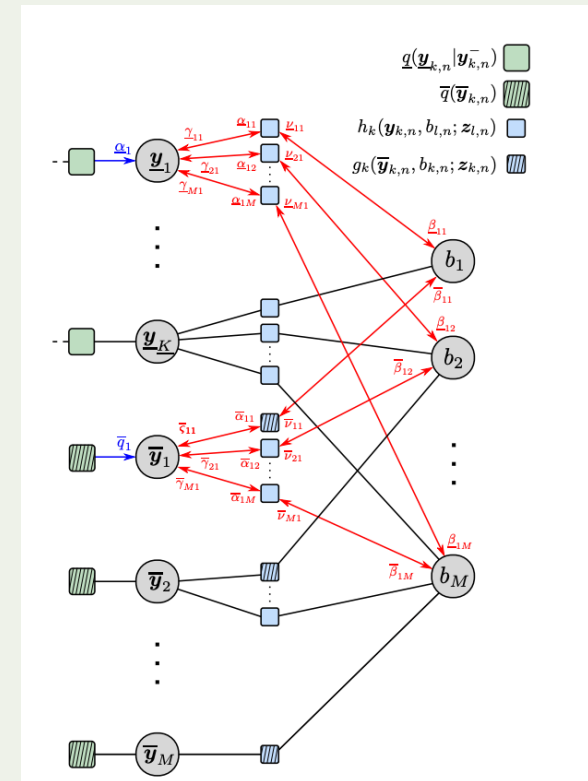
Prediction, birth, and measurement edges

$\underline{q}$: legacy prediction.

$\bar{q}$: new-object unary.

The anchor and later edges encode the canonical labeling.

Red annotations identify BP messages on this graph.



Original Scenario 4 graph. Its $h / \bar{g}$ labels follow the source figure.

Derivation for GrBP

GrBP uses grouped measurements: external clustering fixes one partition.

Data association then decides which target explains each **whole group**.

Introduce one newborn candidate per group.

01 · OBSERVE
Raw detections $\mathbf{Z}$



02 · PRECLUSTER
Fixed partition $\mathcal{P} = \mathcal{C}(\mathbf{Z})$



03 · ASSOCIATE
Groups $\mathbf{Z}^G \rightarrow$ target choices

Points, groups, and targets are different counts.

m raw detections form n^g disjoint groups.

n^t legacy components carry predicted beliefs.

Every component has kinematics, extent, and existence;
 $s = (\underline{x}, \underline{E})$ is shorthand.

$$\mathbf{Z}^G = \{\mathbf{Z}^1, \dots, \mathbf{Z}^{n^g}\}, \quad m = \sum_{j=1}^{n^g} |\mathbf{Z}^j|.$$

$$s = (\underline{x}, \underline{E}), \quad \underline{\mathbf{y}}^p = (\underline{s}^p, \underline{r}^p),$$

$$f(\underline{\mathbf{Y}}) = \prod_{p=1}^{n^t} f(\underline{\mathbf{y}}^p).$$

Two variables record the same claim.

$a^p = j$: legacy target p claims group j .

$b^j = p$: group j comes from legacy target p .

$b^j = 0$ leaves two alternatives:
newborn or clutter.

$$a^p \in \{0, \dots, n^g\}, \quad b^j \in \{0, \dots, n^t\},$$

$$a^p = j \iff b^j = p.$$

$$\psi(\mathbf{a}, \mathbf{b}) = \prod_{p=1}^{n^t} \prod_{j=1}^{n^g} \Psi_{p,j}(a^p, b^j),$$

$$\Psi_{p,j}(a, b) = \mathbf{1}[(a = j) \iff (b = p)].$$

Read the counts from the assignments.

Detected legacy targets, existing newborns, and clutter partition the groups on every feasible hypothesis.

$$\begin{aligned}\mathbb{D} &= \{p : a^p \neq 0\}, & \mathbb{N} &= \{j : \bar{r}^j = 1\}, \\ \mathbb{C} &= \{1, \dots, n^g\} \setminus (\{a^p : p \in \mathbb{D}\} \cup \mathbb{N}), \\ n^d &= |\mathbb{D}|, & n^n &= |\mathbb{N}|, & n^c &= n^g - n^d - n^n.\end{aligned}$$

Separate the prior, allocation, and likelihood.

Condition throughout on the fixed partition $\mathcal{P}$. Its notation is suppressed in the equations.

$$\begin{aligned} f(\underline{\mathbf{Y}}, \overline{\mathbf{Y}}, \mathbf{a}, \mathbf{b}, \mathbf{Z}^G) &= f(\underline{\mathbf{Y}}) \psi(\mathbf{a}, \mathbf{b}) \\ &\quad \times f(\mathbf{a}, \mathbf{b}, \overline{\mathbf{Y}} \mid \underline{\mathbf{Y}}) \\ &\quad \times f(\mathbf{Z}^G \mid \underline{\mathbf{Y}}, \overline{\mathbf{Y}}, \mathbf{a}, \mathbf{b}). \end{aligned}$$

Clutter groups have their own Poisson mean.

μ_g : mean number of clutter **groups**.

μ_n : mean number of **detected newborns**.

Generally $\mu_g \neq \mu_c$: a clutter group can contain several detections.

$$f(n^c) \approx \frac{\mu_g^{n^c} e^{-\mu_g}}{n^c!},$$

$$f(n^n) = \frac{\mu_n^{n^n} e^{-\mu_n}}{n^n!}.$$

The allocation factorials cancel the count factorials.

Choose which unclaimed groups are newborns.

Assign detected legacy targets to distinct group slots.

Multiply by the two Poisson count laws; the remaining denominator is $n^g!$.

$$\begin{aligned} \binom{n^c + n^n}{n^n}^{-1} &= \frac{n^c! n^n!}{(n^c + n^n)!}, \\ \frac{1}{n^g P_{n^d}} &= \frac{(n^c + n^n)!}{n^g!}, \\ \text{product} &= \frac{n^c! n^n!}{n^g!}. \end{aligned}$$

Keep the existence branches and newborn densities.

B collects newborn state densities; f_d is normalized. The nonempty legacy count mass enters the likelihood next.

$$f(\mathbf{a}, \mathbf{b}, \overline{\mathbf{Y}} \mid \underline{\mathbf{Y}}) \approx \frac{\mu_g^{n^c} e^{-\mu_g} \mu_n^{n^n} e^{-\mu_n}}{n^g!} B(\overline{\mathcal{X}} \mid \overline{\mathbf{r}}) \\ \times \prod_{p \in \mathbb{D}} \underline{r}^p \\ \times \prod_{p \notin \mathbb{D}} \left[1 - \underline{r}^p \left(1 - e^{-\mu_m(\underline{s}^p)} \right) \right].$$

$$B(\overline{\mathcal{X}} \mid \overline{\mathbf{r}}) = \prod_{j \in \mathbb{N}} f_n(\overline{s}^j) \prod_{j \notin \mathbb{N}} f_d(\overline{s}^j)$$

A detected newborn uses a nonempty-group likelihood.

A legacy group uses the extended-target Poisson likelihood.

A newborn is detected by definition, so condition on a nonzero count.

Keep one exponential per target, not one per point.

$$L_L(\mathbf{Z}^j \mid s) = e^{-\mu_m(s)} \prod_{\mathbf{z} \in \mathbf{Z}^j} \mu_m(s) f(\mathbf{z} \mid s).$$

$$L_N(\mathbf{Z}^j \mid s) = \frac{L_L(\mathbf{Z}^j \mid s)}{1 - e^{-\mu_m(s)}}, \quad \mathbf{Z}^j \neq \emptyset.$$

Multiply the group likelihoods by source.

Conditioned on a feasible association, the approximate model treats groups as independent.

$$f(\mathbf{Z}^G \mid \underline{\mathbf{Y}}, \overline{\mathbf{Y}}, \mathbf{a}, \mathbf{b}) = \prod_{p \in \mathbb{D}} L_L(\mathbf{Z}^{a^p} \mid \underline{\mathbf{s}}^p) \\ \times \prod_{j \in \mathbb{N}} L_N(\mathbf{Z}^j \mid \overline{\mathbf{s}}^j) \prod_{c \in \mathbb{C}} F_c(\mathbf{Z}^c).$$

Divide each target group by its clutter explanation.

The product of all group clutter densities is a baseline for the entire scan.

Redistributing $\mu_g^{n^c}$ leaves **one** $1/\mu_g$ for every target-generated group.

$$F_c(\mathbf{Z}^j) = \prod_{\mathbf{z} \in \mathbf{Z}^j} f_c(\mathbf{z}),$$

$$\prod_{j=1}^{n^g} F_c(\mathbf{Z}^j) = \prod_{i=1}^m f_c(\mathbf{z}^i).$$

$$\mu_g^{n^c} = \mu_g^{n^g} \prod_{p \in \mathbb{D}} \mu_g^{-1} \prod_{j \in \mathbb{N}} \mu_g^{-1}.$$

Separate the constant and define the group ratio.

C is fixed for the observed grouped scan and fixed model parameters.

$R_j(s)$ abbreviates the per-point likelihood ratios inside group j .

$$C(\mu_g, \mu_n, \mathbf{Z}^G) = \frac{\mu_g^{n_g} e^{-\mu_g} e^{-\mu_n}}{n_g!} \prod_{i=1}^m f_c(\mathbf{z}^i).$$

$$R_j(s) = \prod_{\mathbf{z} \in \mathbf{Z}^j} \frac{\mu_m(s) f(\mathbf{z} \mid s)}{f_c(\mathbf{z})}.$$

A legacy target has four local cases.

An existing target can claim a group or generate no detections.

A nonexistent target can only make a null claim.

The predicted prior remains a separate factor.

$$\underline{l}((s, r), a; \mathbf{Z}^G) = \begin{cases} \frac{e^{-\mu_m(s)}}{\mu_g} R_a(s), & a \neq 0, r = 1, \\ e^{-\mu_m(s)}, & a = 0, r = 1, \\ 1, & a = 0, r = 0, \\ 0, & a \neq 0, r = 0. \end{cases}$$

An unclaimed group can create one newborn.

$b^j = 0$: newborn and clutter compete.

$b^j \neq 0$: a legacy target already owns the group, so the newborn cannot exist.

The absent candidate carries dummy density f_d .

$$\bar{l}((s, r), b; \mathbf{Z}^j) = \begin{cases} \frac{\mu_n f_n(s) e^{-\mu_m(s)}}{\mu_g [1 - e^{-\mu_m(s)}]} R_j(s), & b = 0, r = 1, \\ 0, & b \neq 0, r = 1, \\ f_d(s), & r = 0. \end{cases}$$

The grouped joint PDF becomes a product of local factors.

Predicted priors $\times$ legacy factors $\times$ newborn factors $\times$ group-association consistency.

$$\begin{aligned}
 f(\underline{\mathbf{Y}}, \overline{\mathbf{Y}}, \mathbf{a}, \mathbf{b}, \mathbf{Z}^G) &\approx C \psi(\mathbf{a}, \mathbf{b}) \\
 &\times \prod_{p=1}^{n^t} \left[f(\underline{\mathbf{y}}^p) \underline{l}(\underline{\mathbf{y}}^p, \mathbf{a}^p; \mathbf{Z}^G) \right] \\
 &\times \prod_{j=1}^{n^g} \overline{l}(\overline{\mathbf{y}}^j, \mathbf{b}^j; \mathbf{Z}^j).
 \end{aligned}$$

Association cannot repair a fixed partition.

S4: infer assignments of individual detections; several points can choose the same target.

GrBP: infer whole-group assignments after external clustering.

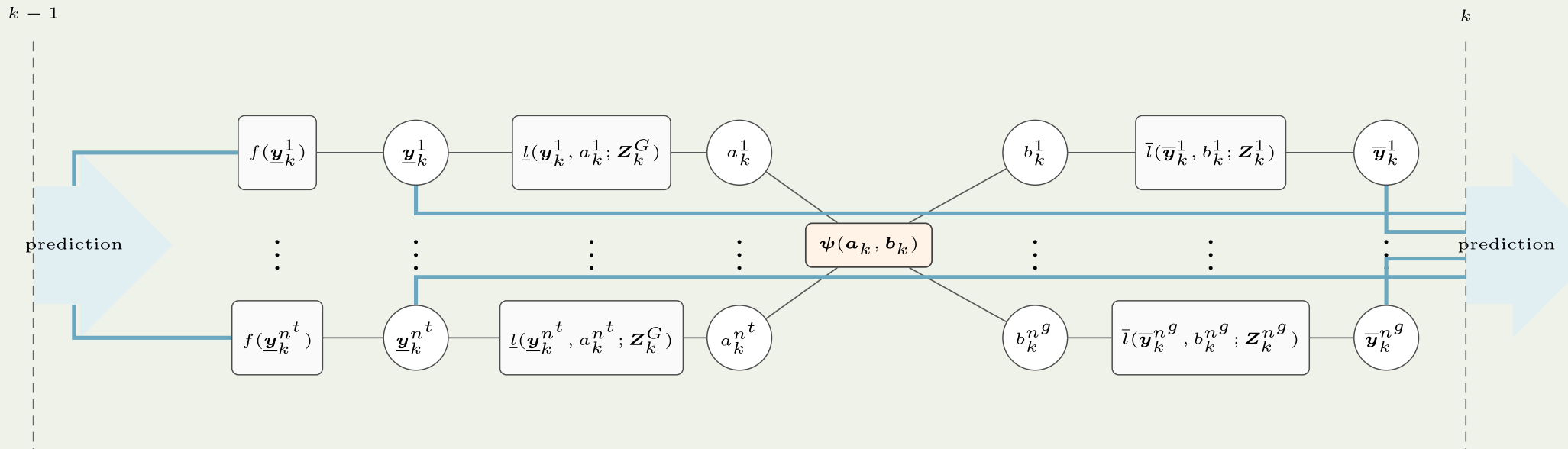
**One group per target.
One source per group.**

Target splits and cross-target merges are outside this model.

The number of groups is not the inferred number of targets.

Factor graph for GrBP

Legacy chains and group-indexed newborn chains meet at $\psi(\mathbf{a}_k, \mathbf{b}_k)$.



Circles: variables · Boxes: factors · Blue: prediction between scans. The central factor collects all pairwise association-consistency factors.

From raw measurements to fixed groups

Unknown object count:

introduce potential objects and existence.

Multiple points per object: use measurement-oriented assignments and Poisson counts.

GrBP: precluster once, then associate whole groups.

Case	Association	Coupling
S1	Target-oriented	Global exclusivity
S2	Target-oriented	Exclusivity + new objects
S3	Measurement-oriented	Object–point factors
S4	Measurement-oriented	Legacy + anchor + later edges
GrBP	Group-level	Exclusive target–group factors

Check the tiny cases first.

1. Keep ordered-vector and finite-set conventions consistent.
2. Retain every state-dependent factor.
3. Verify counts and normalization by enumeration.

1. Keep $\mu_n f_n$ in new-object factors.
2. Enforce the anchor condition and include every later edge.
3. Compare approximate BP marginals against exact inference on small scans.

The page, the scans, and the primary papers

R1 · Point-object models

Meyer et al., “Message passing algorithms for scalable multitarget tracking,” Proceedings of the IEEE, 2018.

R2 · Extended-object model

Granström, Baum & Reuter, “Extended Object Tracking: Introduction, Overview and Applications.”

R3 · Extended objects with uncertain count

Meyer & Williams, “Scalable Detection and Tracking of Geometric Extended Objects,” 2021.

The companion page contains the full derivations and original handwritten scans for S1–S4, plus the derivation for GrBP.